

Robustness of Deep-Learning Dose Prediction in H&N Radiotherapy to Clinically Realistic CT Perturbations

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"How much CT degradation can a dose-prediction model tolerate before its errors become clinically visible?"

THRESHOLD CRITERION

A severity level is "clinically visible" when the cohort-mean change in ANY standard DVH criterion, vs each patient's own baseline prediction, exceeds 1.0 Gy.

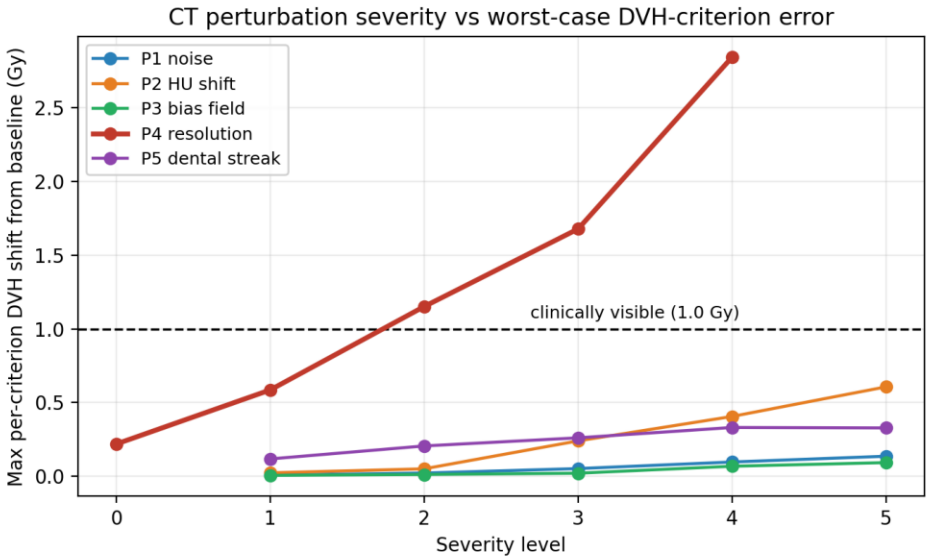
METHODS

3D U-Net + SE blocks, masked MAE, 4x PTV weight (OpenKBP).
40 test patients x 26 CT conditions: baseline + 5 perturbation families x 5 severities, ranges meeting/exceeding ACR QA.
Baseline DVH 2.535, Dose 3.731 Gy.

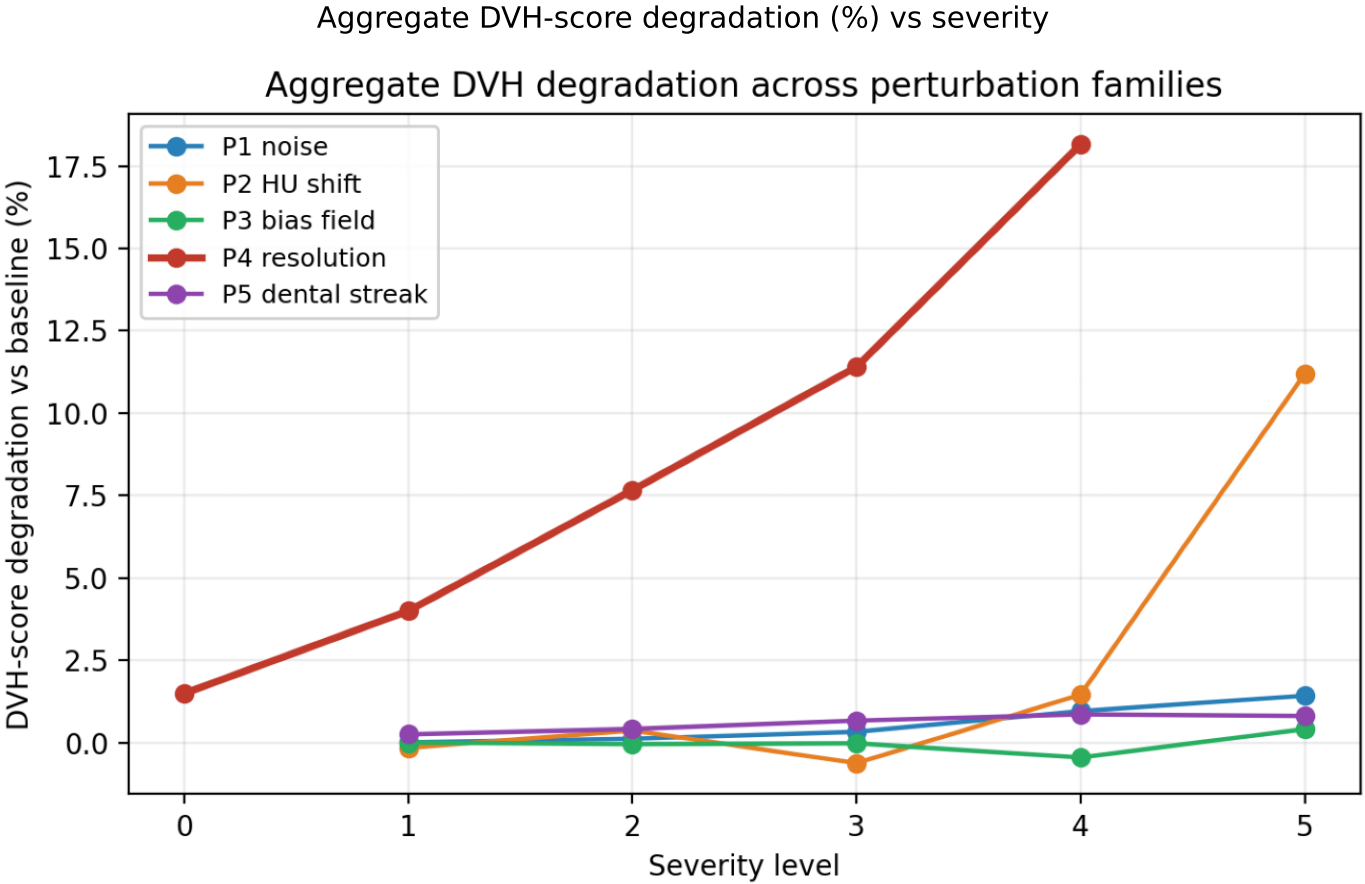
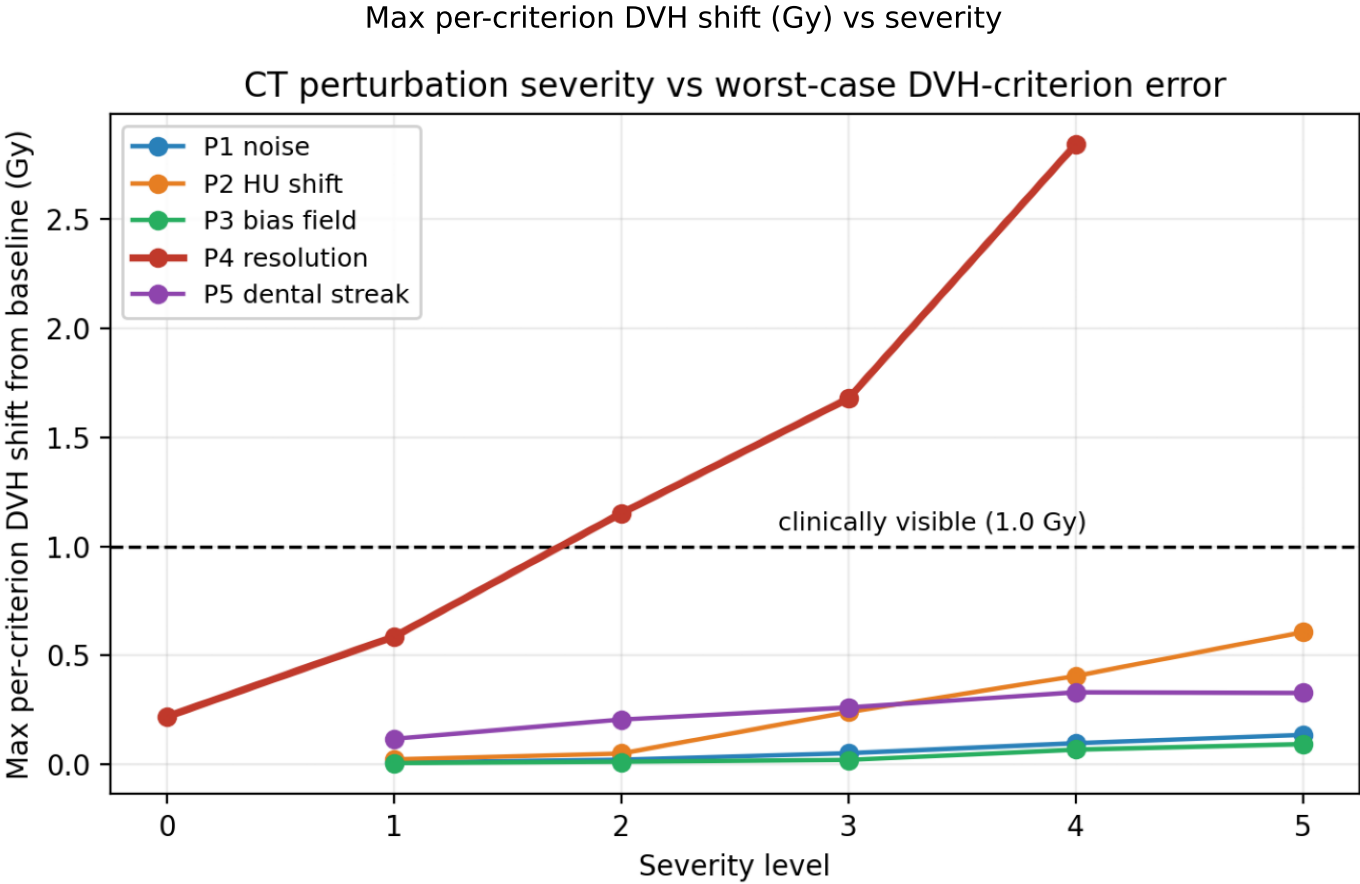
CONCLUSIONS

- Intensity variability (noise, bias field, dental) tolerated through and beyond ACR severities — no criterion > 1 Gy.
- HU calibration shift only bites at implausible ~1000 HU; realistic drift (10–50 HU) is safe.
- Spatial-resolution loss is the dominant failure mode — crosses 1 Gy at L2 (2.0/1.0 vox), within real cross-scanner variation, reaching 2.84 Gy (Larynx D0.1cc) at L4.
- Practical read: deployment QA should prioritize resolution / reconstruction-kernel consistency over intensity calibration.

Family	Range tested	Clinically visible at	Verdict
P1 noise	8/12 HU → 100/160 HU	never (max 0.14 Gy)	Robust
P2 HU shift	5/50 HU → 100/1000 HU	never (max 0.61 Gy)	Robust
P3 bias field	10 HU → 200 HU	never (max 0.09 Gy)	Robust
P4 resolution	0.5/0.25 vox → 4.0/2.0 vox	L2: 1.15 Gy (Larynx D 0.1 cc)	SENSITIVE
P5 dental streak	150HU/8 → 1200/24	never (max 0.33 Gy)	Robust



Severity curves



P4 (resolution) crosses the 1.0 Gy line between L1 and L2 and keeps climbing; every other family stays on the floor.